PRÁCTICA 5.3

Protocolo Centinela: La Inmunización del Código

CICLO FORMATIVO

TÉCNICO EN DAW

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CURSO 2025-2026

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Módulo: Base de Datos

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Tarea 1

Probar los casos normales: 8 y 7.

```
Java
   Caso normal 7
     @Test
    public void testFibonacci() throws Exception {
        System.out.println("fibonacci");
        int n = 7;
        SerieFib instance = new SerieFib();
        int expResult = 13;
        int result = instance.fibonacci(n);
        assertEquals(expResult, result);
        // TODO review the generated test code and remove the
default call to fail.
        //fail("The test case is a prototype.");
    }

Caso normal 8
    @Test
    public void testFibonacci() throws Exception {
        System.out.println("fibonacci");
        int n = 8;
        SerieFib instance = new SerieFib();
        int expResult = 21;
        int result = instance.fibonacci(n);
        assertEquals(expResult, result);
        // TODO review the generated test code and remove the
default call to fail.
        //fail("The test case is a prototype.");
    }
```

```java
public void testFibonacci() throws Exception {
    System.out.println("fibonacci");
    int n = 7;
    SerieFib instance = new SerieFib();
    int expResult = 13;
    int result = instance.fibonacci(n);
    assertEquals(expResult, result);
    // TODO review the generated test code and remove the default call to fail.
    //fail("The test case is a prototype.");
}
```

```
Test (SerieFibTest)
ibonacci
ests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.064 s -- in SerieFibTest

esults:

ests run: 1, Failures: 0, Errors: 0, Skipped: 0

------------------------------------------------------------------------
UILD SUCCESS
------------------------------------------------------------------------
otal time:  2.760 s
inished at: 2026-04-20T16:27:15+02:00
------------------------------------------------------------------------
```

```java
39      /**
40       * Test of fibonacci method, of class SerieFib.
41       */
42      @Test
43      public void testFibonacci() throws Exception {
44          System.out.println("fibonacci");
45          int n = 8;
46          SerieFib instance = new SerieFib();
47          int expResult = 21;
48          int result = instance.fibonacci(n);
49          assertEquals(expResult, result);
50          // TODO review the generated test code and remove the default call to fail.
51          //fail("The test case is a prototype.");
52      }
53
54  }
55
```

g3.entornosss.SerieFibTest > testFibonacci > expResult

Test Results

g3:Entornosss:jar:1.0-SNAPSHOT (Unit)

Tests passed: 100.00 %

The test passed. (0.067 s)

fibonacci

Tarea 2

Probar el caso límite: 1.

```java
Caso límite 1:
@Test
    public void testFibonacci() throws Exception {
        System.out.println("fibonacci");
        int n = 1;
        SerieFib instance = new SerieFib();
        int expResult = 1;
        int result = instance.fibonacci(n);
        assertEquals(expResult, result);
        // TODO review the generated test code and remove the default call
to fail.
        //fail("The test case is a prototype.");
    }


Caso límite 0:
@Test
    public void testFibonacci() throws Exception {
        System.out.println("fibonacci");
        int n = 0;
        SerieFib instance = new SerieFib();
        int expResult = 0;
        int result = instance.fibonacci(n);
        assertEquals(expResult, result);
        // TODO review the generated test code and remove the default call
to fail.
        //fail("The test case is a prototype.");
    }
```

```java
42      @Test
43      public void testFibonacci() throws Exception {
44          System.out.println("fibonacci");
45          int n = 0;
46          SerieFib instance = new SerieFib();
47          int expResult = 0;
48          int result = instance.fibonacci(n);
49          assertEquals(expResult, result);
50          // TODO review the generated test code and remove the default call to fail.
51          //fail("The test case is a prototype.");
52      }
53
54  }
55
```

g3.entornosss.SerieFibTest

Test Results

g3:Entornosss:jar:1.0-SNAPSHOT (Unit)

Tests passed: 100.00 %

The test passed. (0.067 s)

```
fibonacci
```

```java
    @Test
    public void testFibonacci() throws Exception {
        System.out.println("fibonacci");
        int n = 1;
        SerieFib instance = new SerieFib();
        int expResult = 1;
        int result = instance.fibonacci(n);
        assertEquals(expResult, result);
        // TODO review the generated test code and remove the default call to fai
        //fail("The test case is a prototype.");
    }
```

- Test (SerieFibTest)

```
fibonacci
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.074 s -- in SerieFibTest

Results:

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
Total time:  3.096 s
Finished at: 2026-04-20T16:32:15+02:00
------------------------------------------------------------------------
```

Tests (1) finished successfully for project: Ent

Tarea 3

Probar caso de -1:

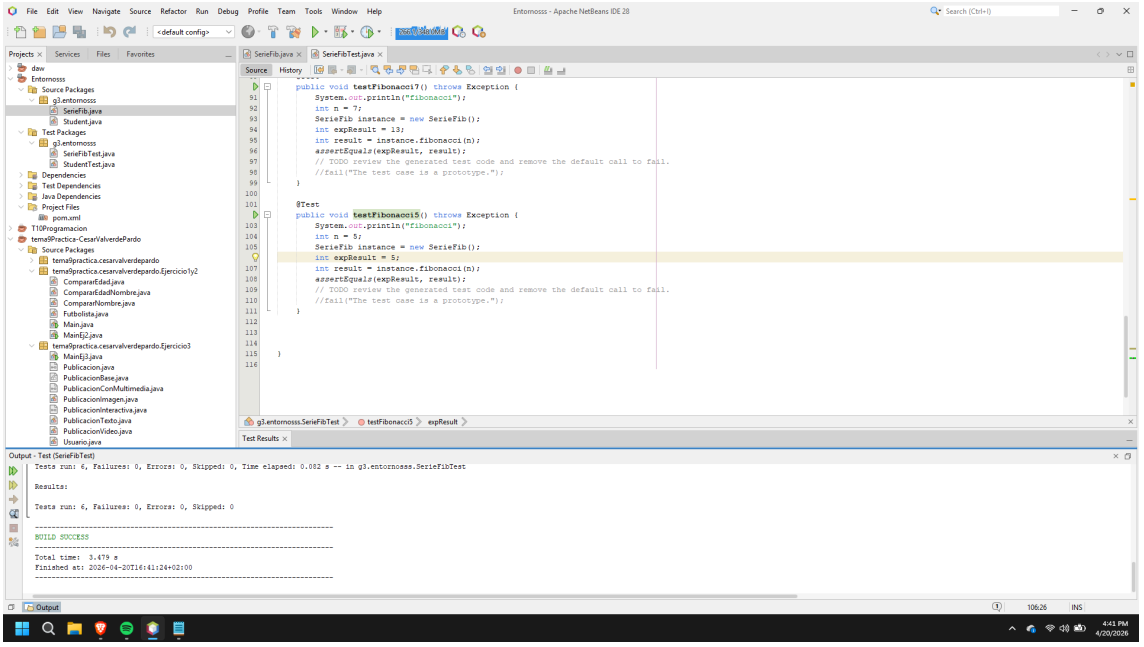
```
Java
Argumento incorrecto -1:
    @Test
    public void testFibonacci() throws Exception {
        SerieFib instance = new SerieFib();
        int numero = -1;
        Exception exception = assertThrows(IllegalArgumentException.class,
() -> instance.fibonacci(numero));
        assertEquals("Fibonacci is not computed for negative numbers",
exception.getMessage());
        // TODO review the generated test code and remove the default call
to fail.
        //fail("The test case is a prototype.");
    }
```

Tarea 4

```
Java
    @Test
    public void testFibonacci() throws Exception {
        SerieFib instance = new SerieFib();
        int numero = -1;
        Exception exception = assertThrows(IllegalArgumentException.class,
() -> instance.fibonacci(numero));
        assertEquals("Fibonacci is not computed for negative numbers",
exception.getMessage());
        // TODO review the generated test code and remove the default call
to fail.
        //fail("The test case is a prototype.");
    }

    @Test
    public void testFibonacci1() throws Exception {
        System.out.println("fibonacci");
```

```
        int n = 1;
        SerieFib instance = new SerieFib();
        int expResult = 1;
        int result = instance.fibonacci(n);
        assertEquals(expResult, result);
        // TODO review the generated test code and remove the default call
to fail.
        //fail("The test case is a prototype.");
    }

    @Test
    public void testFibonacci0() throws Exception {
        System.out.println("fibonacci");
        int n = 0;
        SerieFib instance = new SerieFib();
        int expResult = 0;
        int result = instance.fibonacci(n);
        assertEquals(expResult, result);
        // TODO review the generated test code and remove the default call
to fail.
        //fail("The test case is a prototype.");
    }

    @Test
    public void testFibonacci8() throws Exception {
        System.out.println("fibonacci");
        int n = 8;
        SerieFib instance = new SerieFib();
        int expResult = 21;
        int result = instance.fibonacci(n);
        assertEquals(expResult, result);
        // TODO review the generated test code and remove the default call
to fail.
        //fail("The test case is a prototype.");
    }


    @Test
    public void testFibonacci7() throws Exception {
        System.out.println("fibonacci");
        int n = 7;
        SerieFib instance = new SerieFib();
        int expResult = 13;
        int result = instance.fibonacci(n);
        assertEquals(expResult, result);
```

```
        // TODO review the generated test code and remove the default call
to fail.
        //fail("The test case is a prototype.");
    }

    @Test
    public void testFibonacci5() throws Exception {
        System.out.println("fibonacci");
        int n = 5;
        SerieFib instance = new SerieFib();
        int expResult = 5;t
        int result = instance.fibonacci(n);
        assertEquals(expResult, result);
        // TODO review the generated test code and remove the default call
to fail.
        //fail("The test case is a prototype.");
    }
```



Tarea 5

No hemos tenido ningún problema a la hora de ejecutar tanto haciendo los casos por separado cómo teniendo los casos en el mismo archivo y añadiendo el caso nuevo (5)